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Behavioral/Systems/Cognitive

Signal Propagation and Logic Gating in Networks of Integrate-and-Fire Neurons

Tim P. Vogels and L. F. Abbott
Volen Center for Complex Systems and Department of Biology, Brandeis University, Waltham, Massachusetts 02454-9110

Transmission of signals within the brain is essential for cognitive function, but it is not clear how neural circuits support reliable and accurate signal propagation over a sufficiently large dynamic range. Two modes of propagation have been studied: synfire chains, in which synchronous activity travels through feedforward layers of a neuronal network, and the propagation of fluctuations in firing rate across these layers. In both cases, a sufficient amount of noise, which was added to previous models from an external source, had to be included to support stable propagation. Sparse, randomly connected networks of spiking model neurons can generate chaotic patterns of activity. We investigate whether this activity, which is a more realistic noise source, is sufficient to allow for signal transmission. We find that, for rate-coded signals but not for synfire chains, such networks support robust and accurate signal reproduction through up to six layers if appropriate adjustments are made in synaptic strengths. We investigate the factors affecting transmission and show that multiple signals can propagate simultaneously along different pathways. Using this feature, we show how different types of logic gates can arise within the architecture of the random network through the strengthening of specific synapses.

Key words: rate coding; sensory processing; propagation; integrate-and-fire neurons; network models; synfire chains; logic gates

Introduction

Cognitive processing involves signal propagation through multiple brain regions and the activation of large numbers of specific neurons. Computational approaches are useful for studying the nature and mechanisms of this phenomenon. Two different modes of signal propagation have been proposed. In synfire propagation, the signal is carried by a wave of synchronous neuronal activity within a subset of network neurons (Abeles, 1991; Aertsen et al., 1996; Diesmann et al., 1999). In firing-rate propagation, the wave of activity is an asynchronous elevation in the firing rate of the neurons carrying the signal (van Rossum et al., 2002). In both cases, it has proven surprisingly difficult to construct networks that support stable and robust signal propagation over a sufficiently wide dynamic range to support a significant flow of information. Specifically, adjustments must be made to keep the signal from dying out as it propagates through different subsets of network neurons and also to keep the signal from exploding and spreading to all of the neurons in the network. In addition, large amounts of noise must be included for both forms of propagation, in synfire chains to prevent synchronization from spreading across the network (Diesmann et al., 1999; Aviel et al., 2003; Mehring et al., 2003), and in firing-rate propagation to prevent synchronization of the signal-carrying neurons themselves (van Rossum et al., 2002; Litvak et al., 2003; Reyes, 2003).

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Correspondence should be addressed to Tim P. Vogels, Volen Center for Complex Systems, MS 013, Brandeis University, Waltham, MA 02454. E-mail: vogels@brandeis.edu.

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Noise was included in early studies of signal propagation (Diesmann et al., 1999; van Rossum et al., 2002) by injecting random current into all of the neurons of the network. This corresponds to a source of noise that is external to the network. Using an external noise source has the advantage that the level of noise can be adjusted easily until a level appropriate for signal propagation is found. The propagating signal and noise source are also independent, so they do not interfere with each other. However, an external source of noise is not biologically realistic. Although neurons in real neural circuits show considerable irregularity and variability in their firing (Burns and Webb, 1976; Dean, 1981; Softky and Koch, 1993; Holt et al., 1996; Anderson et al., 2000), the source of this "noisy" activity is internal, not external.

A number of studies have shown that networks of sparsely connected spiking model neurons can produce highly irregular, chaotic activity without any external source of noise (van Vreeswijk and Sompolinsky, 1996, 1998; Brunel, 2000; Mehring et al., 2003; Lerchner et al., 2004; Vogels et al., 2005). These networks allow for the study of signal propagation without the unrealistic injection of external noise. This has been done in the case of synfire chains (Aviel et al., 2003; Mehring et al., 2003), but not for firing-rate propagation. In studies of synfire propagation in irregularly firing networks, a number of complications arose because of interactions between the propagating signal and the ongoing background activity of the network. Signal propagation can be achieved in such networks, but the signals can produce network-wide shock waves that can subsequently silence the background activity of the network (Aviel et al., 2003; Mehring et al., 2003). In light of these complications, we feel it is also important to consider whether firing-rate coded signals can propagate through spontaneously active networks. We must determine

E01 核心缺口：宽动态范围上的可靠传播机制不明
作者报告｜作者在摘要第一句就把研究缺口说清楚：问题不是信号能否传递，而是在足够大的动态范围上可靠、准确地传递的机制未知。这一措辞界定了本文的目标是可行性而非最优化。

E02 两种传播模式的定义
作者报告｜synfire chain 与 firing-rate 传播的操作性定义，两者是本文并列检验的对象。引用时必须保持这一区分，因为本文最重要的结论恰是这两者在同一网络中不可兼得。

E20 只调整强度，不增删连接
作者报告｜计算单元的实现手段被限定为突触强度调整。这与 P2 那类通过改变 E/I 平衡来打开传输的做法是不同机制家族，不应混引。

E06 第二项主张：由突触增强产生逻辑门
作者报告｜作者的措辞是「arise within the architecture」，即在既有随机架构内产生，而非从零构建。引用这一主张时应同时给出其依赖手工增强与预写搜索条件的前提。

E04 对外部噪声源的方法学批评
作者报告｜这是全文的问题设定支点。早期研究靠注入随机电流提供噪声，作者判定其在生物学上不现实，因此必须换成网络内部产生的活动——这一替换是本文全部工作的动机。

E03 已有方法的共同困难：动态范围不够宽
作者报告｜作者对既有工作的评价：两种模式都难做到稳定、稳健且动态范围宽到能支撑显著信息流。这说明本文的贡献在于把动态范围做宽，而不是首次让信号传过去。

E17 层间连接准则：至少三条突触
作者报告｜层规模与连接门限的具体数值。三条这一门限直接决定了后文传播只用到 3 条通路突触、且信噪比先天偏低（见 E30）。

whether such networks generate sufficient internal "noise" to support propagation, because, unlike the external noise case, the level of noise cannot be adjusted; it is fixed by the network. In addition, we must examine whether the propagating signal interferes with the ongoing background activity enough to disrupt its role as a noise source. We also study synfire propagation in the same networks.

Signal propagation has been studied primarily in feedforward networks, either standing alone or embedded into larger networks. To study both rate-code and synfire transmission through a network with self-sustained activity, we do not embed a feedforward structure into the random network used to generate background activity. Instead, we feed a signal into a selected set of network neurons, find the most likely path of transmission through the existing network, and then modify the properties of neurons and synapses along this candidate pathway to determine whether propagation can be achieved. Thus, we are studying signal propagation within the architecture of a sparsely and randomly connected network, not an artificial feedforward structure.

Although faithful signal propagation is a necessary condition for cognitive processing, by itself it does not represent any type of computational process. Once we have established signal propagation in the networks we study, we find and reinforce subcircuits within the existing architecture that provide computational functions. Specifically, we show how logic gates, switches, and memory units can be formed from multiple interacting signal propagating pathways.

Materials and Methods

Neuron model. The network we study is composed of 10,000 leaky integrate-and-fire neurons. Each integrate-and-fire neuron is characterized by a time constant, $\tau = 20$ ms, and a resting membrane potential, $V_{\text{rest}} = -60$ mV. Whenever the membrane potential crosses a spiking threshold of -50 mV, an action potential is generated and the membrane potential is reset to the resting potential, where it remains clamped for a 5 ms refractory period. To set the scale for currents and conductances in the model, we use a membrane resistance of 100 MΩ.

We model the synaptic connections between these neurons in two different ways, as currents and as conductances, resulting in either current-based (CUBA) or conductance-based (COBA) models. For the CUBA model, the subthreshold membrane potential obeys the following equation:

$$\tau \frac{dV}{dt} = (V_{\text{rest}} - V) + g_{\text{ex}}(E_{\text{ex}} - V_{\text{rest}}) + g_{\text{inh}}(E_{\text{inh}} - V_{\text{rest}}), \quad (1)$$

whereas, in the COBA model, the membrane voltages are calculated as follows:

$$\tau \frac{dV}{dt} = (V_{\text{rest}} - V) + g_{\text{ex}}(E_{\text{ex}} - V) + g_{\text{inh}}(E_{\text{inh}} - V). \quad (2)$$

Reversal potentials are $E_{\text{ex}} = 0$ mV and $E_{\text{inh}} = -80$ mV. The synaptic conductances (or effective synaptic conductance in the case of the CUBA model) g_{ex} and g_{inh} are expressed in units of the resting membrane conductance.

Neurons in the network are either excitatory or inhibitory. When a neuron fires, the appropriate synaptic variable of its postsynaptic targets are increased, $g_{\text{ex}} \rightarrow g_{\text{ex}} + \Delta g_{\text{ex}}$ for an excitatory presynaptic neuron and $g_{\text{inh}} \rightarrow g_{\text{inh}} + \Delta g_{\text{inh}}$ for an inhibitory presynaptic neuron. Otherwise, these parameters obey the following equations:

$$\tau_{\text{ex}} \frac{dg_{\text{ex}}}{dt} = -g_{\text{ex}} \quad (3)$$

and

$$\tau_{\text{inh}} \frac{dg_{\text{inh}}}{dt} = -g_{\text{inh}}, \quad (4)$$

with synaptic time constants $\tau_{\text{ex}} = 5$ ms and $\tau_{\text{inh}} = 10$ ms. In most cases, rather than reporting the values of Δg_{ex} and Δg_{inh}, which are the synaptic strengths, we report the resulting EPSP and IPSP sizes. These are obtained within the active network from spike-triggered average membrane potentials of postsynaptic neurons after spikes evoked within individual network neurons. The integration time step for our simulations is 0.1 ms.

Network architecture. To create self-sustained, asynchronous background activity, we chose a 4:1 ratio of excitatory to inhibitory neurons in a network of 10,000 cells and connected them to each other randomly with a connection probability of 2%. This value was chosen as a compromise between the higher connection probabilities found for neighboring neurons in cortex and the lower values for neurons separated by distance. Our results should apply to models with connection probabilities up to ~10%. Except along the signaling pathway (see below), all excitatory synapses took the same strength, as did all of the inhibitory synapses. These two sets of strengths were adjusted to allow asynchronous activity within the network.

Input signals. To test signal propagation through the network, we generated a set of Poisson input spike trains with a firing rate $r_0(t)$. These form a 0th layer, L_0, that provides input to the network. Input spikes generated by the Poisson process in layer 0 were fed into a layer 1 subpopulation of the network neurons, labeled L_1, by increasing their excitatory synaptic conductances by $g_{\text{ex}} \rightarrow g_{\text{ex}} + \Delta g_0$ whenever they received an input spike from the layer 0 source. The synaptic strength Δg_0 was tuned so that the firing rates of the layer 1 neurons reproduce the input signal, that is, they track the input firing rate $r_0(t)$. To analyze propagation, we fed various signals $r_0(t)$ into the network. At first, square-wave pulses at 180 Hz lasting for 30 ms were used to assess propagation. Then, constant input firing rates were used to study the elevation of firing rates across different propagation layers L_i for $i = 1, 2, \ldots, 6$. Finally, the accuracy of signal propagation was examined by constructing $r_0(t)$ from Gaussian-distributed white noise low-pass filtered at 50 ms and half-wave rectified (van Rossum et al., 2002). In addition, the temporal properties of signal propagation were analyzed using input rates that varied sinusoidally at different frequencies, by measuring the onset delay in each layer for a propagated constant stimulus, and by studying propagation for synchronous stimuli. The results we obtain by injecting layer 0 spikes into the network could also be obtained by injecting current into the layer 1 neurons.

Signaling pathways. Signal propagation is investigated along specific pathways found within the full network. We do not change the network architecture by, for example, adding a feedforward pathway to the network, and we do not require all-to-all connectivity between pathway layers. Instead, we look for existing pathways already in the network. To do this, we look for the most likely candidates for neurons that will be affected by layer 1 activity. A potential signal-propagation pathway is a series of layers of neurons that are connected to each other in a feedforward manner more highly than average because of random fluctuations in network connectivity. By tracing these most likely candidates through the existing network, we uncover a potential signal-propagation pathway.

Specifically, we identify candidate propagation pathways in the following manner. First, we choose 33 neurons randomly as layer 1 neurons that received input from layer 0. Then, by searching the networks, we find 33 neurons, each of which receives three or more synapses from the neurons in layer 1 (although the condition is three or more, the number is three ~90% of the time). These define layer 2. A third layer of neurons is constructed in a similar manner by finding 33 network neurons that receive three or more synaptic connections from the neurons of layer 2. An additional requirement on layer 3 neurons is that they must not receive any direct connections from layer 1. This prevents "short-circuiting" of the multilayered propagation pathway. This procedure is continued with layer i, for $i = 4, 5, 6$, defined as a set of 33 neurons, each of which receives at least three synapses from the neurons of layer $i - 1$, and receives no synapses from layers $i - 2, i - 3, \ldots, 1$. In the networks we study, probabilities of connections between neurons are given by binomial distributions. This allows us to compute the expected numbers of neurons in each of the layers of a propagation pathway. The expected number of neurons satisfying the conditions for layer 2 is 250, but the "no

E14 网络架构的三个参数
作者报告｜背景网络的具体设定。作者紧接着说明该连接概率是皮层中邻近神经元与远距神经元之间的折中，并称结论应适用于连接概率到 10% 的模型——这是作者给出的推广范围声明。

E07 方法学选择：不嵌入前馈结构
作者报告｜决定本文结论适用范围的关键选择：传播通路必须来自随机网络本身。由此可知本文证明的是通路存在且可被打开，而不是通路会自行形成。

E08 研究对象的自我定位
作者报告｜作者用否定句强调自己不是在人工前馈结构里做传播，这一限定使本文的结论不能直接套用到分层的皮层前馈模型上。

E18 首轮测试信号的形式
作者报告｜测试按三个层次推进：方波脉冲定性、常数速率定量、滤波白噪声检真伪。这说明作者先确认传播存在，再考察传输质量，而非一步到位。

E63 最优因子取决于信号性质
作者报告｜作者指出 13 倍这一数字不是普适常数，而取决于信号本身的时序统计。这条限定使「最优突触因子」成为信号性质的函数，跨信号类型引用时必须重新扫描。

E11 神经元模型：单舱室 LIF
作者报告｜本文的证据类型由此确定：全部结论来自单舱室 LIF 网络的数值模拟，不含任何实验数据，也没有生物学细节如树突、离子通道种类或形态。

E13 交叉验证的声明
作者报告｜重要限定：每个定性效应在 CUBA 中也出现，但作者的措辞是「most of the results ... are for COBA」，因此定量差异（如图 5c 的最优因子平台只存在于 COBA）并未被交叉验证，引用时不能笼统说两者等价。

E16 结构不被改动，也不要求全连通
作者报告｜两条硬约束：网络架构保持随机，层间不要求全连通。与 C1 那类工作对照时要注意，本文不假设任何解剖上的分层或柱状结构。

short-circuit" condition reduces the number in subsequent layers significantly. For layer 6, we predict only ~10 candidate neurons. These are, of course, mean values. In the actual simulations, we take advantage of fluctuations around the mean (there are 200 ± 14 connections per neuron) that allow for more cells in higher layers. The number 33 was chosen as a compromise between maximizing the number of neurons in each layer and the number of layers that satisfy the prescribed conditions (sometimes for layer 6, or when multiple pathways were identified, <33 neurons were found that satisfied all of the conditions and a smaller layer size was used). Usually there are not enough cells available to support a seventh layer. Increasing the connectivity of the network decreases the number of cells that fulfill the no short-circuit rule, and hence decreases the number of layers. We call the neurons within layer 1–6 pathway neurons and the synapses between them pathway synapses. We control signal propagation along the pathway by modifying the excitability of pathway neurons or the strength of the pathway synapses.

Logic gates. To study signal processing within the network, we search for candidate subcircuits that fulfill conditions for the computation we want to achieve. The circuits we discuss are not constructed by hand, but instead are found within the random architecture of the network. Once a candidate circuit is found, adjustments are made to the synapses within the circuit, but no new connections are made nor are any existing connections eliminated.

To identify a candidate NOT gate, we search for a group of inhibitory cells that is highly connected (at least three synapses) to an output layer of cells spontaneously firing at rates between 30 and 40 Hz. We also seek out a third layer consisting of excitatory cells that have three or more synapses to the inhibitory cell population. The entire candidate gate consists of ~60 neurons. To make this candidate unit functional, excitatory synapses are strengthened by sixfold and inhibitory synapses by twofold.

To identify a neuronal "switch," we search for a regular pathway consisting of three excitatory layers, as described above. Additionally, we found an inhibitory cell population with three or more synapses per cell to the output layer of the excitatory pathway. Driving these inhibitory cells with a population of highly connected excitatory neurons makes it possible to control propagation along the excitatory pathway. To make this candidate unit functional, excitatory synapses are strengthened by 10-fold and inhibitory synapses by 2-fold.

The candidate pathway for an XOR gate uses two reciprocally connected switches that funnel their outputs into a common layer of neurons. In such XOR gates, two input layers drive layers of both excitatory and inhibitory interneurons, with the inhibitory neurons synapsing onto the excitatory interneurons of the other switch. Neurons are chosen to be maximally connected, as discussed in the previous paragraph. Input into one path, thus, drives the output layer while inhibiting propagation along the other path of the gate. A functional XOR gate involves ~220 neurons. To make such a candidate unit functional, excitatory synapses are strengthened by 10-fold and inhibitory synapses by 2-fold.

To find a candidate "flip-flop," two recurrently connected loops of excitatory neurons are identified, with the additional requirement that each loop synapses onto a layer of inhibitory interneurons that in turn contacts the other loop. With sufficiently strengthened synapses, it is possible to maintain an elevated firing rate in one loop while suppressing it in the other, even in the absence of external input. A functional flip-flop involves ~200 neurons. To make this candidate unit functional, excitatory synapses are strengthened by 10-fold and inhibitory synapses by 8-fold. For this network structure, fine-tuning that depends on the actual connectivity of the given network is necessary.

Analysis. To characterize sustained asynchronous network activity, we monitor individual neuronal membrane potentials, the population firing rate (the average of the firing rates across the network), average membrane potentials, and interspike intervals (ISIs). The asynchronous irregular network activity we seek has an approximately constant population firing rate and coefficients of variation (CVs) near 1 for the ISI distributions of the neurons. The ISI CV for a neuron is the ratio of the SD of the ISI distribution to its mean. CV values close to 0 indicate regular spiking patterns, values near 1 indicate irregular spiking, and values >1 indicate, in our simulations, burstiness in the firing pattern, in which a neuron is likely to fire several spikes in a short interval followed by a longer period of silence.

Signal propagation is analyzed by determining how similar the firing rates of the different layers along the propagation pathway are to the layer 0 input rate. To do this, we calculate the firing rates of pathway neurons in 5 ms bins by counting their spikes. We use the notation $r_i(t)$ for the average of the firing rates of all of the neurons in layer i for the time bin centered at time t, and we denote the time average of this firing rate by $\bar{r}_i$. As a measure of similarity of the firing of neurons in layer i to those in layer 0, we compute the correlation coefficient as follows:

$$C_i(\tau) = \frac{\langle (r_0(t)-\bar{r}_0)(r_i(t+\tau)-\bar{r}_i)\rangle_t}{\sqrt{\langle (r_0(t)-\bar{r}_0)^2\rangle_t \langle (r_i(t+\tau)-\bar{r}_i)^2\rangle_t}}, \quad (5)$$

where the brackets denote an average over t values. We use the activity of layer 0 as a reference, rather than layer 1, to distinguish signal transmission from propagation of fluctuations arising spontaneously in layer 1. Signal propagation up to layer i is characterized by reporting the maximum value of $C_i(\tau)$, which we call the similarity. The value of τ, where this maximum occurs, is the propagation delay for that layer. To test for the propagation of synfire chains, synchronous spikes are fed into layer 1. Spike number conservation across layers is used as an estimate for propagation success.

Results

We are interested in studying signal propagation within a network that sustains, by itself, ongoing activity characterized by irregular firing that is asynchronous across the network. This pattern of firing generates the noise needed for propagation. We begin by discussing the nature and properties of this irregularly and asynchronously firing network, and then address how signals propagate through it.

Sustained activity

Previous studies have shown that large networks of sparsely connected integrate-and-fire model neurons can sustain irregular asynchronous activity (van Vreeswijk and Sompolinsky, 1996, 1998; Amit and Brunel, 1997; Brunel, 2000; Mehring et al., 2003; Lerchner et al., 2004; Vogels et al., 2005). In most of these studies, synaptic connections between neurons were simulated by injecting an exponentially decaying pulse of current into the postsynaptic neuron whenever the presynaptic neuron fired an action potential. We call such models CUBA for "current based." An alternative is to transiently change the conductance of the postsynaptic neuron after a presynaptic action potential. We call these models COBA for "conductance based." Because biological synapses involve conductance changes and not current injection, it might appear that COBA models would inevitably be more realistic than CUBA models, but this is not necessarily the case. The individual neurons in large network simulations are typically represented by single compartments that must approximate the full dendritic structure of the real neurons being modeled. Depending on the placement of synapses on the dendritic tree, a CUBA or a COBA single-compartment model, or a mixture of the two, may be a more accurate representation. For this reason, we consider both CUBA and COBA models in our studies. Most of the results we show are for COBA models, but every effect reported was also reproduced in an analogous CUBA model.

Large, sparsely coupled networks display self-sustained irregular asynchronous firing of the type needed to support signal propagation if their parameters are adjusted appropriately. For our purposes, three conditions need to be met: sustained activity (for at least as long as it takes to examine signal propagation), relatively low firing rates, and ISI CVs near 1. To find such activity, we performed a parameter search of a 10,000 neuron COBA

E09 上限条件：通常凑不出第七层
作者报告｜六层是本文架构下的上限附近而非任意深度。引用「六层传播」时必须带上这个限定，否则会高估该结论的可推广性。

E10 层数上限随连通度下降
作者报告｜这一条把「六层」明确绑定到本文的无短路搜索规则：连通度升高反而使可用神经元减少。若把本文模型移入更稠密的架构，层数上限会进一步下降。

E21 分析指标：相似度与传播时延
作者报告｜指标设计上的一个关键细节：用 layer 0 而非 layer 1 作参考，才能把「信号确实被传入」与「layer 1 里的自发涨落被放大传播」分开。这是本文相似度数值可信的前提。

E72 噪声的双重作用
作者报告｜作者对计算单元的总体评价：噪声是必要的（否则环路会同步锁死）但也是失败来源。这条结论直接对应本清单所关心的「关门时瞬态是否泄漏」，在本文中表现为随机失败而非可控泄漏。

E19 计算单元同样来自搜索
作者报告｜作者强调逻辑门不是手工搭建的。但要准确理解：它们是按预先写定的连接条件**搜出来的**，因此这一声明支持「无需新建连接」，不支持「电路自行涌现」。

E05 核心主张：rate 可行而 synfire 不可
作者报告｜摘要给出的是一条**互斥**结论而非两个独立的成功：rate 编码可行、synfire 不可行，条件是做了恰当的突触强度调整。忽略「but not for synfire chains」就会误读本文的乐观程度。

E15 通路搜索取代通路构建
作者报告｜核心方法：先找最可能的既有路径，再只修改路径上的性质。这一顺序决定了本文是在既有结构中做选择，而不是让结构为信号让路。

E12 同时做电流型与电导型两种模型
作者报告｜作者并不假定电导型更真实，而是两种都做，理由在紧邻的前文：单舱室近似哪种更准确取决于突触在真实树突上的位置。这个选择使 COBA 与 CUBA 的比较成为有意的对照设计。

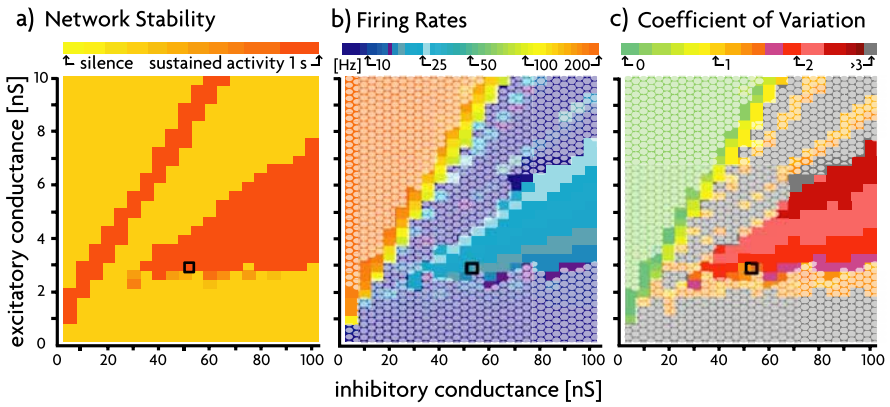


Figure 1. Parameter search. Excitatory and inhibitory conductances refer to the parameters Δg_{ex} and Δg_{inh} converted to nanosiemens assuming a resting neuronal membrane conductance of 100 MΩ. The black square shows the parameter values used in all subsequent COBA simulations. ***a***, Duration of network activity. Parameter pairs in which network activity was sustained over the length of the simulation (1000 ms) are colored in orange. Pairs leading to silent networks are shown in yellow and the same regions are denoted by a white mesh in ***b*** and ***c***. ***b***, Average network firing rates. Firing rates in configurations with sustained activity range from 8 to 200 Hz. ***c***, Average CV of ISIs. CVs range from 0 (very regular) to 3 (very bursty) over the range in which activity was sustained.

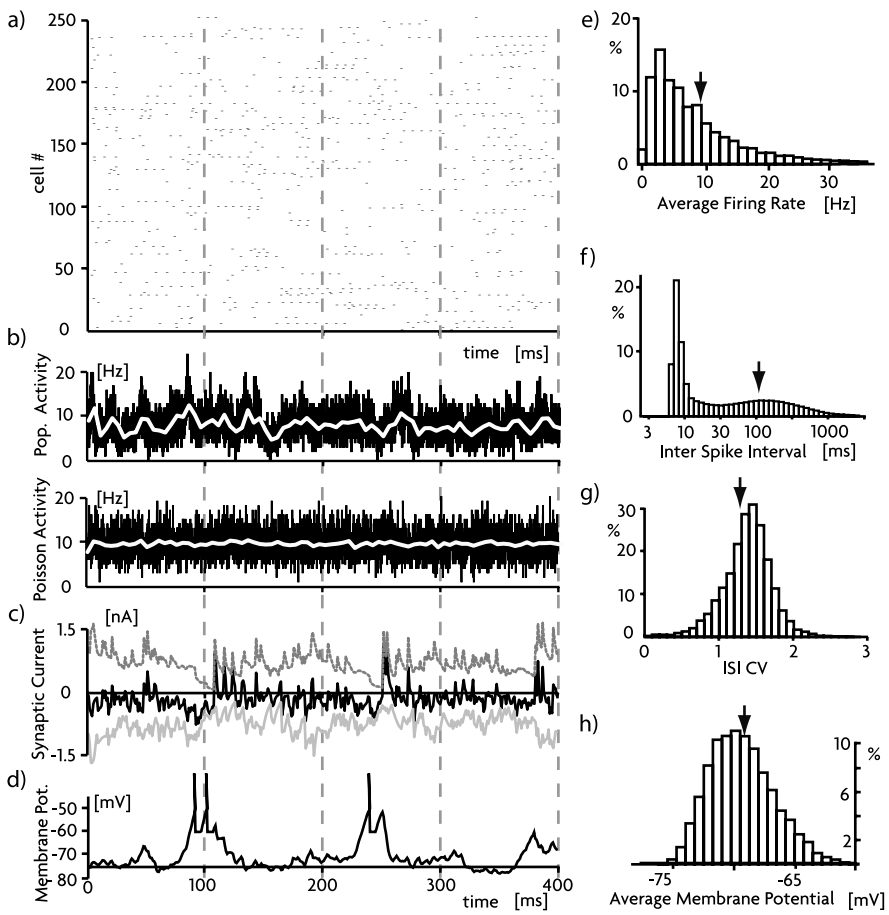


Figure 2. Background activity in a COBA model. ***a***, Spike raster for a sample set of 250 neurons over a simulated time of 400 ms. ***b***, Average firing rate of the entire population and of a Poisson train. The black trace shows the rate computed from 0.1 ms bins, and the white trace shows the same activity computed using 5 ms bins. The top panel is computed from the network, and the bottom panel, for comparison, from an equivalent number of Poisson processes with a 5 ms refractory period producing spikes at the same rate as the network. ***c***, Membrane currents of a randomly chosen neuron. Inhibitory currents are in dark gray, excitatory ones are in light gray, and the total synaptic current is shown in solid black. ***d***, Membrane potential of a randomly chosen neuron. ***e***, Distribution of firing rates of the network neurons. ***f***, Distribution of ISIs of the network neurons. ***g***, Distribution of CVs of ISIs of the network neurons. ***h***, Distribution of average membrane potentials of the network neurons. ***e–h***, The arrow marks the mean of the distribution. Pop., Population; Pot., potential.

model with 2% connectivity by systematically varying the strengths of its excitatory and inhibitory synapses (Fig. 1). Figure 1*a* indicates how long activity was sustained in the network after a brief period of initial stimulation, Figure 1*b* gives the average firing rate of the network neurons (for the period while the activity lasted in the case of transiently active networks), and Figure 1*c* shows the ISI CV averaged over all network neurons. The black square in these figures denotes a state that satisfies the requirements listed above (actually, the "sustained" activity in the COBA network lasts for as long as our simulations run, but it will not last indefinitely) with conductances that correspond, on average, to 1 mV EPSPs and −2.6 mV IPSPs within the active network. This state is used for the remaining figures in which a COBA model appears.

With the parameters described by the black square in Figure 1, the network stays active with an average firing rate of 9 Hz and an average ISI CV of 1.2 (Fig. 2*e*,*g*). Network activity for these parameters is asynchronous. Neither the raster plot of the spike times of 250 sample cells (Fig. 2*a*) nor the average population activity (Fig. 2*b*, top) shows obvious temporal structure, and autocorrelations also do not reveal any obvious firing patterns (data not shown). To quantify the asynchronous nature of the firing, we compared the population firing rate obtained from the network (Fig. 2*b*, top) with equivalent fire rates derived from a Poisson process generating spikes at the same rate (Fig. 2*b*, bottom). The Poisson spikes produce a smoother firing rate than the network, indicating that there is some temporal structure in the network activity, but the overall levels of the fluctuations are similar. For example, the variance of the firing rate measured in 0.1 ms bins for the network is 1.5 times that of the Poisson train.

Excitatory and inhibitory currents are balanced and tend to cancel each other (Fig. 2*c*), keeping the average membrane potential of the network neurons at −70 mV (Fig. 2*h*). The sample single-neuron membrane potential trace in Figure 2*d* shows that spiking is irregular but tends to come in bursts. This is further revealed by the distribution of ISIs (Fig. 2*f*), which has a peak at small intervals reflecting this bursting and a second local maximum at a larger interval indicating the typical interburst interval.

It is possible to obtain activity similar, but not identical, to that shown for the COBA model in Figure 2 using a CUBA model (Fig. 3*a*) with 0.25 mV EPSPs and

E22 背景态参数的选取标准
作者报告｜参数搜索的判据是作者预先列出的三项要求（活动持续、发放率较低、ISI CV 接近 1）。因此 COBA 参数是被挑选出来的最优点，不是从网络中自动涌现的属性。

E23 COBA 背景态的定量表现
作者报告｜选定背景态的两个核心数字：平均 9 Hz、平均 ISI CV 1.2。CV 略大于 1 而非等于 1，作者在后文指出这对应轻微的成簇发放，CUBA 模型则没有这一性质。

E24 异步性的量化证据
作者报告｜作者用与同率泊松列车的方差比（1.5 倍）来量化「网络比泊松更不光滑、但涨落量级相当」。这是本文对异步性的少数几个量化指标之一，也是判断噪声强度是否合适的直接依据。

E25 背景态的 E/I 平衡
作者报告｜背景态处于 E/I 平衡，膜电位被稳定在约 −70 mV。这一点是理解后文增益调制失败的关键前提：E/I 平衡使得同时增强兴奋与抑制输入几乎不改变平均放电率。

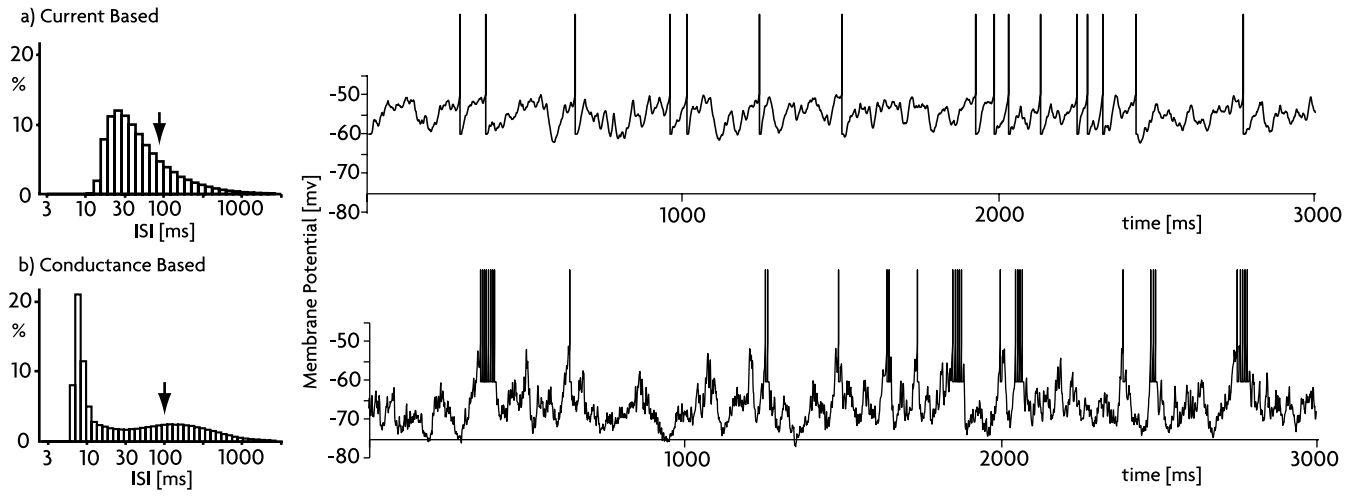


Figure 3. Differences in the background activity of the CUBA (***a***) and COBA (***b***) models. Distribution of ISIs in each model plotted on a semilog scale is shown (left). The arrow marks the mean of each distribution. Membrane potentials of a randomly chosen neuron in each model are shown (right). Note that the two models are operating in different parameter regimes (see Results).

−2.25 mV IPSPs. Although both the CUBA (Fig. 3*a*) and COBA (Fig. 3*b*) models shown have the same average firing rate and both display irregular asynchronous activity, they differ in two important respects. First, spiking in the CUBA model shows much less bursting than in the COBA model. This is revealed by the absence of both a short-interval peak and a long-interval interburst peak in the ISI distribution of the CUBA model (Fig. 3*a*,*b*, compare left panels) as well as being apparent in the sample membrane potential traces (Fig. 3*a*,*b*, right panels). Interspike intervals in the CUBA model approximately follow a Poisson distribution, except for small ISIs that are suppressed by the imposed refractory period. Another important distinction is that the CUBA model includes a constant current injected into all of the neurons of the network sufficient to depolarize them by 11 mV. This causes the average membrane potential in the active network to be −55 mV, whereas the average membrane potential for the COBA models is −70 mV. Without this additional input, the CUBA network does not sustain activity. In contrast, the COBA model sustains activity without any injected current. Note that the two network configurations shown differ significantly in the sizes of their postsynaptic potentials. It is possible to build a COBA model very similar to the CUBA model shown in Figure 3*a*, but only a COBA model can produce reasonable results in the high synaptic-strength regimen shown in Figure 3*b*.

The ability of the COBA model to generate and sustain asynchronous activity without current injection is attributable to the voltage dependence of the EPSP and IPSP amplitudes for conductance-based synapses (Kuhn et al., 2004; Kumar et al., 2005; Moreno-Bote and Parga, 2005). In the CUBA model, EPSP and IPSP amplitudes are voltage independent, but in the COBA model, driving-force effects cause EPSPs to shrink and IPSPs to grow with increased depolarization, or EPSPs to grow and IPSPs to shrink with increased hyperpolarization. This provides a stabilization mechanism that holds the membrane potential much steadier than it is in the CUBA model. The resulting buffering protects the network against excessively low or high firing rates and creates enough stability to sustain activity.

Signal propagation

We study signal propagation by randomly choosing a set of 33 neurons, called layer 1 or L_1, and injecting an input signal into them (see Materials and Methods). For Figure 4, the input was a square pulse of presynaptic activity from layer 0 (shown as the blue raster under the axis in Fig. 4*b*–*e*), large enough to significantly elevate firing in the L_1 neurons (Figs. 4*b*–*e*, red raster). We then examined the effect that this layer 1 firing had on other network neurons. We chose to monitor the neurons most likely to be affected by defining a set of layer 2 neurons that were connected to the layer 1 neurons by at least three synapses. We defined layer 3 neurons as a set connected to the layer 2 neurons by at least three synapses and proceeded in this manner to define six layers of neurons along the potential signal propagation pathway (see Materials and Methods). Figure 4*a* shows the structure of such a pathway, and the activities of the neurons in its different layers are indicated by the different colored rasters in Figure 4*b*–*e*.

If we make no modifications to the synapses or neurons along the candidate pathway in the network, firing-rate signals fail to propagate beyond layer 1 (Fig. 4*b*). The way the network is set up, only three synapses typically extend from layer 1 to a particular neuron in layer 2. In addition to the input from layer 1, each cell of layer 2 receives input from ~140 excitatory and 40 inhibitory neurons. Even accounting for the elevation in the firing rates of layer 1 neurons above the average activity of 10 Hz within the network, the signal from layer 1 represents a small minority of the EPSPs that a layer 2 neuron receives. With such a signal-to-noise ratio, it is not surprising that signals fail to propagate.

We tried several different strategies to promote propagation: depolarizing neurons along the pathway, increasing their response gain, or increasing the strengths of the excitatory synapses between pathway neurons. Depolarization of the pathway neurons leads to a general increase in their firing rates, but the signal fails to propagate beyond layer 1 (Fig. 4*c*). Increasing the gain of neurons, which is usually thought of as changing the slope of the input/output function to modulate intrinsic excitability, is, in the case of the integrate-and-fire neurons we use, equivalent to increasing the strengths of all of the synapses (excitatory and inhibitory) that they receive. This sensitizes the pathway neurons to all of their inputs, not just the signal-carrying ones. In this case, the average firing rate does not increase significantly, because the depolarizing and hyperpolarizing membrane currents remain balanced. Even with gain modulation of pathway neurons, the signal fails to propagate beyond layer 1 (Fig. 4*d*).

Propagation of the firing-rate signal through all six pathway

E26 CUBA 必须外部注入才能自持
作者报告｜CUBA 网络的自持依赖于外部注入的恒流，这与作者批评早期 synfire 研究的理由是同一个批评。作者随后指出，COBA 网络不需要任何注入电流，这正是本文选 COBA 作主模型的原因之一。

E29 基线失败：未改动则信号不过第 1 层
作者报告｜传播实验的对照基线。必须先有这个失败，「增强通路突触有效」才成为有意义的结论，而不仅仅是一个可以成立的现象。

E47 机制：网络是差的时间边沿检测器
作者报告｜作者用一句话概括频率响应的成因：方波响应随层推进不断展宽，因此快速变化会被相邻响应互相淹没。这一诊断解释了 E46 而不依赖任何拟合。

E30 信噪比：3 条突触对 180 个输入
作者报告｜作者解释基线为何失败：携带信号的突触只占该神经元全部输入的很小一部分。数字对照是每个第 2 层神经元还从约 140 个兴奋与 40 个抑制性神经元接收输入，因此本文的传播难度被稀疏连接显著抬高。

Ｅ２７　　ＣＯＢＡ　无需注入即可自持
作者报告｜ＣＯＢＡ　的自持性来自电导机制而非外部偏置，这使　ＣＯＢＡ　网络成为「完全内部噪声源」的实例。需要注意作者同时说明只有　ＣＯＢＡ　能在高突触强度区域给出合理结果，即两者工作在不同参数区。

Ｅ３１　　去极化干预失败
作者报告｜第一种被检验并被否定的干预：把通路神经元抬高工作点只让它们普遍更活跃，并不选择性传递信号。这是「活动增加不等于信号传递」的直接证据。

Ｅ２８　　稳定化机制的物理解释
作者报告｜作者把　ＣＯＢＡ　的自持归因于驱动力效应（去极化时　ＥＰＳＰ　缩小、ＩＰＳＰ　增大）。这是本文对「电导型突触带来何种好处」给出的机制解释，也是步骤　４　中　ＣＯＢＡ　归因争议的起点。

Ｅ３２　　增益调制的机制：增强全部输入
作者报告｜作者对增益调制的操作性定义：在　ＬＩＦ　模型中它等价于同时提高该神经元接收的全部突触强度。这一定义是理解其失败原因的关键，也是引用本文讨论增益调制时必须附带的定义。

Ｅ３３　　增益调制干预失败
作者报告｜本文最有引用价值的负结果：在稀疏随机网络中，沿途的增益调制不能替代通路特异性的突触增强。机制上是因为增益调制对背景输入与信号输入一视同仁，且　Ｅ／Ｉ　平衡使平均放电率几乎不变。

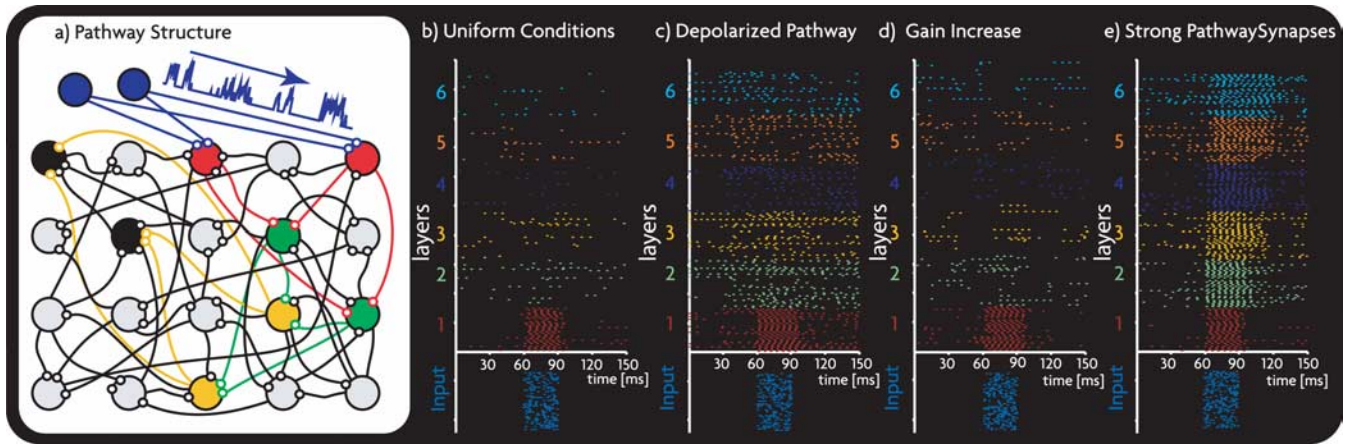


Figure 4. Signal propagation. ***a***, Network diagram showing the layers of a candidate pathway. Input (blue) is fed into the network through strong synapses onto layer 1 neurons (red). In this and the following diagrams, layers 1–6 are indicated by the colors green, yellow, dark blue, orange, and light blue, respectively. The gray-filled circles denote nonpathway neurons of the network. For this figure, layer 0 activity consists of a 30 ms pulse of activity at ~180 Hz. ***b***, In a network with uniform excitatory and inhibitory synaptic strengths and neuronal parameters, no propagation occurs. ***c***, Depolarization of pathway neurons by 15 mV fails to induce propagation, although firing rates in all affected cells increase. ***d***, Gain increase of pathway neurons. Because gain modulation maintains the excitatory/inhibitory balance, firing rates do not increase significantly, but activity fails to propagate. ***e***, Strengthening of pathway synapses by ~10-fold results in signal propagation.

layers without synchronization can be achieved by increasing the strengths of the synapses that carry the signal from one layer of pathway neurons to the next by ~10-fold (Fig. 4*e*). Other network synapses not along the propagation pathway are not altered. We define the synapse enhancement factor, or synapse factor, for short, as the ratio of the strength of excitatory pathway synapses to excitatory synapses not along the pathway minus 1. In other words, for a synapse enhancement factor x, the strength of an excitatory pathway synapse is $1 + x$ times that of an excitatory nonpathway synapse. The result in Figure 4*e* indicates that the spontaneous activity within the network is sufficient to support signal propagation.

To further investigate the effects of strengthening pathway synapses, we fed a constant input signal from layer 0 into the network. Figure 5, *a* and *b*, shows the firing rates induced in layers 1–6 (different colored lines) of a COBA network by this constant input for two different layer 0 rates (*a*, 50 Hz; *b*, 170 Hz). The background firing rate of neurons not along the propagation pathway is also shown (lowest blue line). The induced layer 1 firing rate (red curve) is close to the layer 0 firing rate and is approximately independent of the synapse enhancement factor, because the synapses from layer 0 to layer 1 are held fixed. For low values of pathway synaptic strength (synapse enhancement factors not much greater than 0), the firing rates in layers 2–6 are unaffected by the layer 0 input, and they remain near background levels. As the strength of pathway synapses is increased, layers 2–6 increasingly respond to the input. Eventually, at the point in Figure 5, *a* and *b*, at which all of the different colored curves cross (other than the curve for the background rate), all of the layers respond to the input by firing at approximately the same rate. Such agreement in the firing rates indicates faithful rate propagation through the pathway layers of the network. We call the synaptic factor at which this crossing occurs the optimal synapse enhancement factor. If the synapse enhancement is further increased beyond the optimal value, spontaneous as well as evoked activity is transmitted along the pathway, which drowns out the signal.

We extracted optimal synapse enhancement factors for different layer 0 firing rates from sets of curves similar to those in Figure 5, *a* and *b*, for both COBA and CUBA models. For the COBA model, the curve of optimal synapse factor versus firing

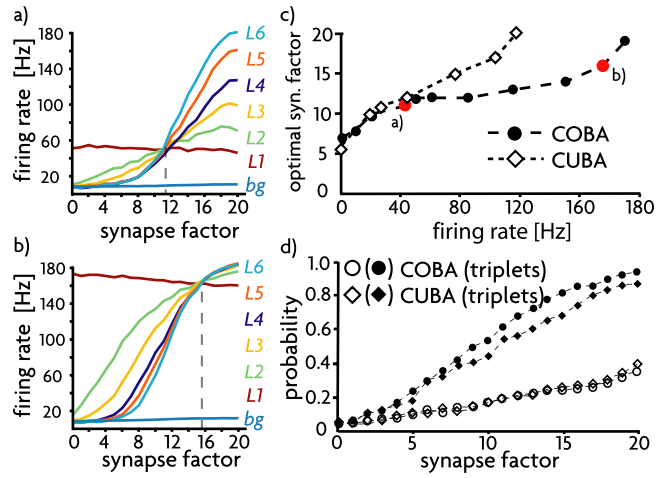


Figure 5. Optimal synaptic enhancement. ***a***, ***b***, Average firing rates of layer 1 (red), 2 (green), 3 (yellow), 4 (dark blue), 5 (orange), and 6 (light blue) in response to a constant layer 0 rate of 50 Hz (***a***) or 170 Hz (***b***). The background rate of nonpathway neurons is indicated by the straight black line at the bottom. The ratio of the strength of pathway synapses to nonpathway synapses is 1 plus the synapse factor. The optimal synapse enhancement factor is indicated by the vertical dashed line. ***c***, The optimal synapse enhancement factors in COBA and CUBA models for different layer 0 firing rates. The examples shown in ***a*** and ***b*** are filled in red. ***d***, The probability of a postsynaptic spike within a 5 ms window of a presynaptic single spike (open symbol) or a synchronous triplet of presynaptic spikes (solid symbols), plotted as a function of the synapse factor for both COBA and CUBA models.

rate has a long plateau at a value of 12, stretching from a layer 0 firing rate of 30 Hz to 160 Hz (Fig. 5*c*, solid symbols). This means that a wide range of firing rates can be transmitted across a pathway using a fixed factor for the enhancement of pathway synapses. When the layer 0 rate is too low, the signal gets lost in the 9 Hz background firing of the network, and network neurons cannot fire faster than 200 Hz because of the imposed refractory period. Thus, transmission occurs pretty much over the entire range in which signal propagation could have been expected, using a single value of synapse enhancement. The plateau value of the optimal enhancement was ~13-fold stronger than the nonpathway network synapses. In the active network, these synapses create average EPSPs of 8 mV. In the CUBA model (Fig. 5*c*, open symbols), no plateau exists, indicating that the range of firing

E34 成功的干预：只增强通路突触
作者报告｜全文核心结论的落点：内部自发活动足以支撑 rate 传播。注意作者只声称「sufficient to support」，没有声称稳健或普适。

E35 过度增强会淹没信号
作者报告｜双侧失败的存在：不是越强越好。这一条使「存在最优突触因子」成为结构性结论而非调参巧合，也说明本文的门控强度是一个窄窗口。

E37 最优因子换算成相对强度
作者报告｜13 倍是把工作点换算到实验可比较尺度的关键数字。它同时也说明本工作区的实现依赖人工给定的通路，而非网络中自然涌现的强连接。

E38 换算成 EPSP 幅度
作者报告｜13 倍增强对应活跃网络中约 8 mV 的平均 EPSP。作者在 Discussion 中称这一幅度「相对偏大但并非闻所未闻」并引 Song 等 2005 作为合理性参照。

E36 最优突触因子的速率平台
作者报告｜最优因子不随输入速率变化（30–160 Hz 内恒为 12），说明存在一个宽的稳定工作点而不只是一个临界点。这是本文结果最具工程价值的一条，也是 CUBA 模型所缺少的。

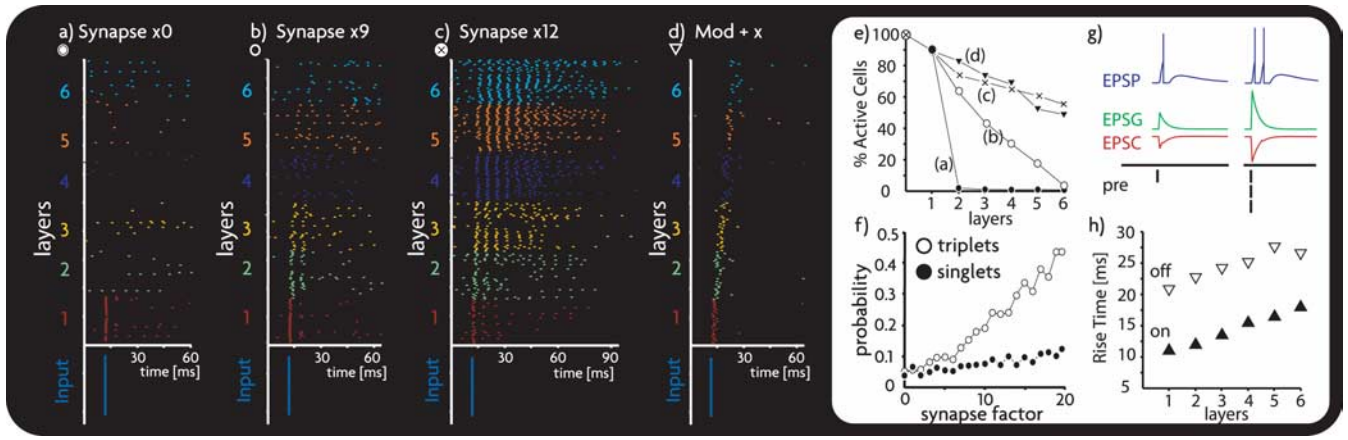


Figure 6. A study of synfire waves in response to a synchronous signal fed into layer 1. ***a***, No propagation occurs in the unaltered network. ***b***, For synapse factor 9, the signal propagates up to layer 3 but then gets washed out by spontaneous activity. ***c***, For synapse factor 12, the signal evokes a response in all layers, but packet length increases with every layer because of secondary spiking. ***d***, When the pathway neurons are gain modulated to decrease their responsiveness by 10-fold and the synapse factor is 30, the difference between pathway and nonpathway synapses can be increased enough to propagate a synfire wave through six layers. Mod + x, Gain modulated and synapses strengthened. ***e***, Average number of active cells in the first propagated wave front of a synfire event, plotted for the different cases described in ***a–d***. ***f***, Probability of evoking a secondary spike within 5 ms of the end of the refractory period of the first spike, plotted for presynaptic singlets (solid) and triplets (open). Each point is calculated from 5000 stimuli delivered to randomly chosen neurons. ***g***, Secondary spikes are evoked when the postsynaptic conductance is large. ***h***, Rise-time delays for on (solid) and off (open) signals consisting of pulses between 0 and 100 Hz at synapse factor 12. Higher layers have slower rise times, because they are affected by the rise times of their precursors as well.

rates that can be transmitted in the CUBA model is much more restricted than in the COBA model for a fixed enhancement of pathway synapses.

As an alternative measure of synaptic strength, we determined the average probability that a spike from a neuron in the propagation pathway evoked a spike within 5 ms in one of its postsynaptic targets in the next layer of the pathway (Fig. 5*d*). Over the range of synapse enhancement factors considered, this increased from near 0 to 0.4. Synchronous triplets of presynaptic inputs increase the probability of postsynaptic spiking by approximately threefold, similar to a linear summation of three separate events.

In addition to firing-rate propagation, we tested for synfire propagation within the network by evoking synchronous events in layer 0, with submillisecond precision. In the unaltered network, a synfire wave failed to propagate beyond layer 1 (Fig. 6*a*), as expected from earlier results, in which many more than three "spike units" were needed for successful synfire propagation (Diesmann et al., 1999). Strengthening pathway synapses by a factor of 9 produces pulses of activity in the first few layers, but the synfire wave ultimately gets dispersed by the background activity of the network (Fig. 6*b*). At a synaptic enhancement of 12, propagation occurs but a different problem arises (Fig. 6*c*). The initial tight synfire packet expands as it propagates through the layers. This phenomenon is somewhat reminiscent of what has been described as synfire explosions (Mehring et al., 2003), a surge of high activity that holds the entire network hostage and subsequently silences it completely. Explosions like these do not occur in our network, probably because of its sparse and random connectivity. The activity stays confined to the pathway, but the duration of the synfire packet increases across the layers.

It is possible to achieve synfire propagation in the network by combining a negative gain modulation along the pathway (i.e., weakening all synapses onto pathway neurons) with a strengthening of pathways synapses (Fig. 6*d*). This decreases the effect of the background activity along the pathway but keeps the rest of the network from becoming synchronized with the synfire pulse by maintaining high noise levels for nonpathway neurons. Figure 6*e* summarizes the results of Figure 6*a–d* in terms of the percentage of spiking neurons within each layer during the propagation (or nonpropagation) of the leading edge of the synfire wave.

The problem of lengthening synfire packets seen in Figure 6*c* is a general property of network propagation. The same kind of lengthening can be seen for a firing-rate pulse in Figure 4*e*. The probability to evoke a postsynaptic spike with a synchronous triplet increases linearly with the number of synchronous presynaptic spikes (Fig. 5*d*), but the total number of evoked spikes grows faster because second and third postsynaptic spikes are evoked (Fig. 6*f*,*g*). With sufficiently strong synapses, a single synchronous volley evokes a train of two or three synchronous spikes in the next layer, which in turn evoke an even longer train in subsequent layers (Fig. 6*c*). Packet lengthening limits the frequency response for firing-rate propagation (but to realistic levels) (Fig. 7*g*), but it is fatal to synfire propagation, because this requires the faithful reproduction of a signal containing substantial high-frequency components. The same synaptic enhancement factor that allows the leading edge of the synfire packet to propagate causes the trailing edge to lag more and more from one layer to the next. Figure 6*h* shows the rise time delays for on and off signals from 0 to 100 Hz and vice versa as solid and open symbols, respectively. The longer rise time delay for off-signals is attributable to this packet-lengthening effect.

We further tested firing-rate propagation by generating a time-varying firing rate in layer 0 from filtered white noise (see Materials and Methods) (Fig. 7*a*). Figure 7*b* shows average firing rates in three of the six pathway layers. The firing rates in different layers clearly track each other, but occasionally the network transmits "ghost signals" [as for example at $t \approx 500$ ms (Fig. 7*b*)] that arise spontaneously. Figure 7*c* shows that the induced fluctuations along the pathway do not significantly modify the overall firing rate of the entire network.

To quantify the quality of signal transmission, we calculated the correlation coefficients between the firing rate of layer 0 and the average firing rates of the neurons in the six network pathway layers (see Materials and Methods). The resulting six correlation coefficients depend on the time delay at which the firing rate correlation is computed, peaking at a value that gives the time delay for propagation up to each layer (Fig. 7*d*, different colored lines). These delays vary from 0 in layer 1 to 20 ms in layer 6. In Figure 7*e–h*, we plot the value of the correlation coefficient at its maximum, which we call the similarity. Figure 7, *e* and *f*, reiter-

E42 包长增长对 synfire 是致命的
作者报告｜作者把包长增长确立为网络传播的普遍性质（rate 脉冲中同样可见），并区分其后果：对 rate 只是限制频率响应，对 synfire 则是致命的，因为 synfire 要求忠实重现高频成分。

E40 中等增强：被背景冲散
作者报告｜synfire 的中间情形：能传几层但被背景活动冲散。值得注意的是因子 9 对 rate 传播已经足够，可见两种模式对突触强度的需求不一致。

E44 传播涨落不改变全网活动
作者报告｜作者检查了传播是否会破坏背景活动这一关键担忧（早期 synfire 研究中曾出现全网冲击波并压制背景）。结论是本文设定下不构成问题，但作者也记录了自发产生的 ghost signals。

E41 synfire 成功的必要条件：负增益调制
作者报告｜synfire 传播必须叠加沿通路的负增益调制（减弱通路神经元接收的全部突触），与通路突触增强结合才可行。与步骤 3 中增益调制对 rate 传播无效形成对照，说明增益调制在这里的作用是压制噪声而非放大信号。

E43 逐层累积的传播时延
作者报告｜六层累计约 20 ms 的传播时延，约每层 3–4 ms。这个数字与实际神经回路的时间尺度可比，因此是本文结果中最有生物学参考价值的量之一。

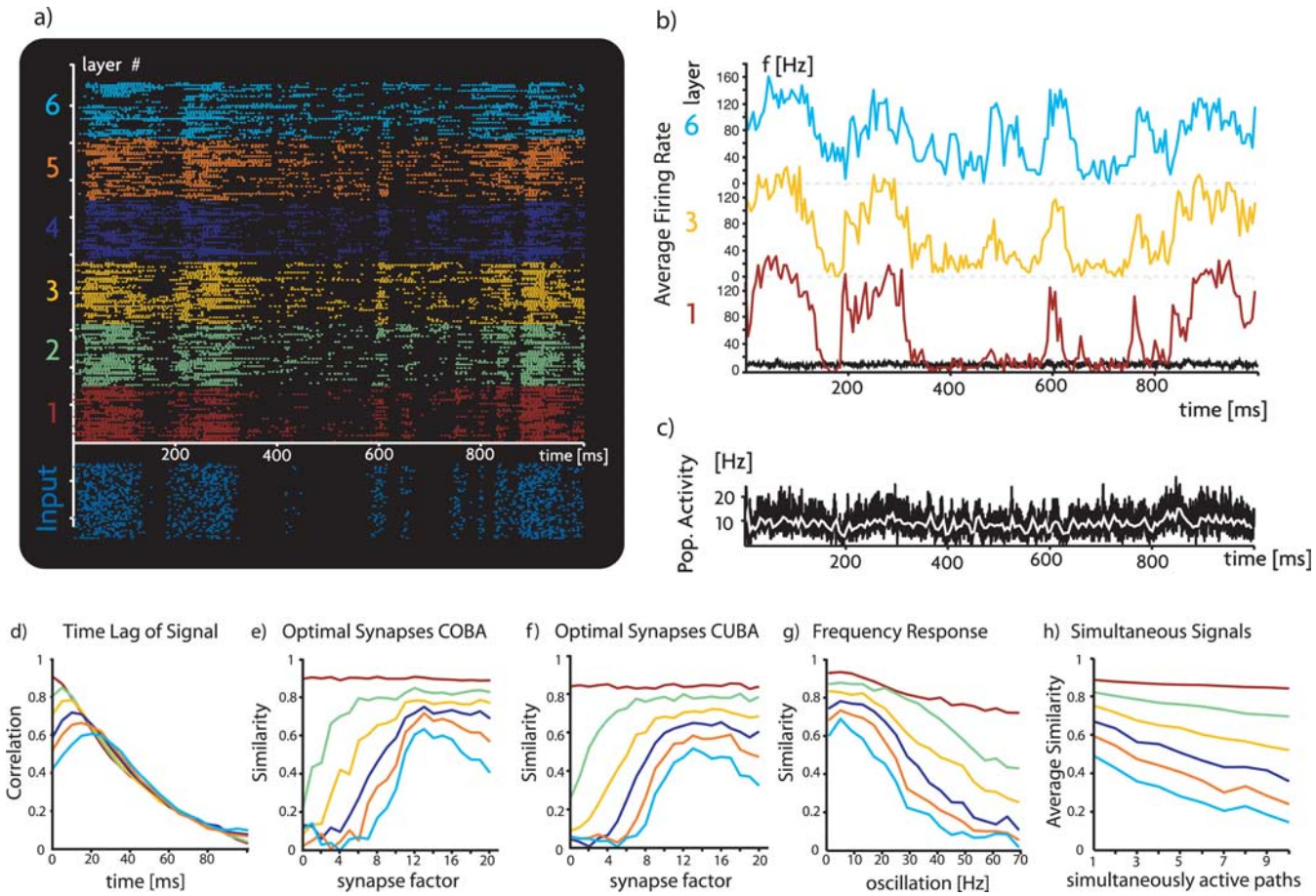


Figure 7. Transmission of time-varying signals in a COBA network. ***a***, Raster of the propagation of a randomly varying layer 0 firing rate through all six layers. ***b***, Average firing rates, calculated in 5 ms bins, for layers 1, 3, and 6 responding to a randomly varying layer 0 rate. ***c***, Average network firing rates (black trace, 0.1 ms bins; white trace, 5 ms bins) are relatively unaffected by the propagating fluctuations. ***d***, Correlations of layer 1 (red), 2 (green), 3 (yellow), 4 (dark blue), 5 (orange), and 6 (light blue) firing rates with the layer 0 rate computed at various time delays. ***e***, Similarity values of layer 1–6 firing rates with the layer 0 rate as a function of the synaptic enhancement factor in the COBA model. Optimal transmission occurs at a synapse factor of 12, corresponding to a 13-fold increase in synaptic strength. ***f***, Same as ***e***, but for the CUBA model. ***g***, Similarity values for layer 0 rates oscillating sinusoidally at different frequencies. ***h***, Similarity values when different numbers of signals propagate through the network along 10 different pathways.

ates the point made in Figure 5 about optimal synapse enhancements and show that the tuning of this factor is most critical for propagation through the later layers of the pathway in both COBA (Fig. 7*e*) and CUBA (Fig. 7*f*) models. In the COBA model with optimally strengthened synapses, the firing rate in layer 6 matches the layer 0 rate with a similarity value of 60%. Similarity values are somewhat smaller in the CUBA model than in the COBA model because of the plateau in the optimal synaptic enhancement curve for the COBA model in Figure 5*c*.

We next tested transmission using sinusoidally varying layer 0 firing rates and found that transmission fidelity depends on signal frequency (Fig. 7*g*). Transmission is most accurate at ~5 Hz, and accuracy falls off rapidly beyond ~20 Hz. The reason for this can be seen in the raster of Figure 4*e* (see also Fig. 6*c*,*h*). The network is a poor temporal edge detector. As discussed above, the duration of the response to a square wave input grows as the signal propagates through the layers, and it is significantly lengthened by layer 6. Figure 6*h* shows that it takes each layer ~15 ms to adjust the layer 6 firing rate from background activity to a steady state of 100 Hz, and even longer to return to the background level. This translates into a maximum change of ~6 Hz/ms. Once the signal changes faster than this, responses start to wash into each other and the signal gets lost. The peak at 5 Hz occurs because oscillations at this frequency best mask spontaneously arising ghost signals. The frequency maximum and fall-off seen in Figure 7*g* match quite well with frequency responses measured in cortical neurons (Movshon et al., 1978; Orban et al., 1985; Hawken et al., 1996).

With 10,000 network neurons and ~200 pathway cells, it is possible to find several different signaling pathways within a network and feed different signals into them simultaneously. Figure 7*h* shows similarity values for the six layers along from 1 to 10 pathways. Although transmission quality drops with multiple pathways because of interference between them, it is still possible to transmit signals at levels greater than the noise along multiple pathways. With this many pathways, the probabilistic nature of network connectivity does not allow us to find 33 neurons that satisfy all of the conditions for layer 6. If we use the average similarity value in layer 5, rather than layer 6, as a criterion for transmission quality to compensate for this, we see that the accuracy of the reproduced spike patterns in layer 5 falls below 50% for four simultaneously active pathways. With ~200 neurons involved in each pathway, this corresponds to using 8% of the network neurons for signal propagation.

Signal processing

Signal propagation is a prerequisite for signal processing, and we now show that with signal propagation accomplished, the net-

E69 与皮层频率响应的定性比对
作者报告｜本文唯一一处与实测数据对照的结果，但属定性比对：作者未报告拟合参数、未做定量误差评估。引用为「与皮层频率响应吻合」时应注明这一性质。

E45 终点相似度 60%
作者报告｜传输质量的终点指标：第 6 层与第 0 层的相似度 60%。作者说明 CUBA 模型更低，并归因于 COBA 才有的最优增强平台。这是「传播成功」与「传播精确」之间差距的量化体现。

E50 多通路并行传输可行但有干扰
作者报告｜作者在 10 条通路上同时传输不同信号，得到「可行但精度下降」的结论。这一条把本文从单通路分析扩展到并行编码，是本文对建模最实用的部分之一。

E46 频率响应：5 Hz 最准，20 Hz 外骤降
作者报告｜传输保真度强烈依赖信号频率，窗口相当窄。引用本文的「信号可以传播」时必须带上这一条，因为多数生理信号包含远超 20 Hz 的成分。

E51 多通路的失效阈值
作者报告｜并行的定量边界：4 条通路同时激活时第 5 层的重建精度跌破 50%。作者说明此时层 6 已凑不满 33 个合格神经元，因此改用第 5 层作为判据——换判据这件事应在引用时说明。

E48 时间常数的定量后果
作者报告｜把频率限制换算成时间常数：第 6 层每层约 15 ms 才能调整到位，回落更久。作者由此给出最大变化率约 6 Hz/ms，即快于此速率的信号开始丢失。

E52 并行传输的资源代价
作者报告｜每条通路约 200 个神经元，4 条并行即占网络 8% 的神经元。这个资源占用比例是评估「在多大网络上做多少路并行」时的实用参考。

E49 5 Hz 峰的来源：掩盖 ghost 信号
作者报告｜作者解释了频率响应峰值的含义：不是因为 5 Hz 传得最「顺」，而是因为该频率的振荡最能掩盖自发产生的虚假信号。这个解释同时解释了为什么峰值不在更高频率。

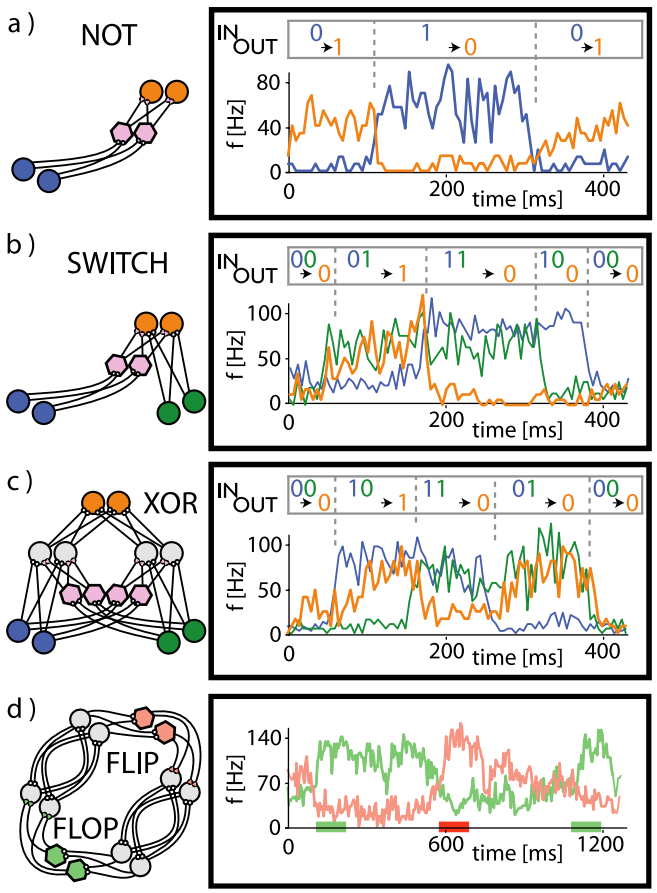


Figure 8. Processing units constructed by synaptically tuning existing network subcircuits. ***a***, NOT gate. ***b***, Switch. ***c***, XOR gate. ***d***, Flip-flop. The left side of each subplot shows the layout of each circuit, with inhibitory neurons drawn as hexagons. On the right, the average firing rates of input and output layers are plotted along with the Boolean interpretation above the traces. In ***d***, external input to the different loops is indicated by color-coded bars along the *x*-axis.

work we are studying can also perform computations. To do this, we identify candidate processing circuits within the existing architecture of the network (by procedures discussed in Materials and Methods), just as we identified candidate pathways previously. Also, as before, we then strengthen synapses within the identified circuit to turn a candidate processing unit into a functioning one. A common feature of all four circuits shown in Figure 8, already discussed for the excitatory pathway in Figure 7*h*, is their relatively sluggish responses, amplified further by the longer inhibitory synaptic time constant.

Figure 8*a* illustrates the structure for a candidate NOT gate, in which a set of inhibitory interneurons controls an upstream layer, so that driving the inhibitory layer silences the otherwise active output layer. In Boolean terms, this represents an output of 1 for input 0 and an output of 0 for input 1. Figure 8*b* shows an extension of the strategy used to trace out the NOT gate, to create a switch that controls propagation along an excitatory pathway. Driving the inhibitory cell population of the circuit impedes any signal propagation along the excitatory pathway. Although the example shown is an on/off switch, the same circuit can be used to modulate a propagating signal in an analog manner by varying the firing rate along the inhibitory pathway (data not shown).

To create an XOR gate (an exclusive OR gate that propagates signals when either one of two pathways is active but not both), two entwined switches synapsing onto the same output layer are identified. When all synapses are strengthened sufficiently, the output layer fires above 60 Hz for single inputs and well below 40 Hz for simultaneous inputs along both paths. Translated into Boolean terms, this represents the output of an XOR gate (Fig. 8*c*).

Figure 8*d* shows the result of strengthening synapses within a candidate flip-flop circuit as a basic memory unit. A flip-flop must maintain high firing rates in one of two recurrent loops, even in the absence of an external signal, while suppressing activity in the other through a set of inhibitory interneurons. When one of the two loops is driven by an external source, the loop sustains its firing rate even after the input is shut off. The flip-flop can reverse its state when the other loop is stimulated. Such activity is seen in Figure 8*d*. Although the circuit can function as a flip-flop, it is not perfect. Because of fluctuations in the background activity, the inhibitory activity from one loop is sometimes insufficient to silence the other. Such a failure is seen ~800 ms into the simulation of Figure 8*d*, at which the flip-flop spontaneously changes its state, thus failing to maintain a memory. Elevated and asynchronous sustained firing rates in such units are possible only because of sufficient background activity, but this activity can cause failures too.

Discussion

We studied signal propagation in two different types of network models, COBA and CUBA. For the CUBA network, we used irregular asynchronous activity as observed in previous work, and for the COBA network, we chose parameters that also created irregular asynchronous but, more importantly, self-sustained activity. In such a network configuration, the size of postsynaptic events is approximately one magnitude larger than in the CUBA network, the average membrane potentials are lower, and the resulting activity has a more burst-like character. In addition, we tested some of our results in the low conductance regimen of the COBA network, with similar outcomes. Both models provide an internal, nonadjustable source of noise sufficiently large to prevent synchronization but not large enough to destroy signals and, thus, support rate-mode signal propagation. Furthermore, interference between the propagating signal and the background activity did not prove problematic. Signal transmission in the COBA model is more accurate, in the sense that a wider range of firing rates can be transmitted across the layers of the signaling pathway without parameter adjustment. This is because of the presence of conductance-based synapses, rather than the fact that the CUBA and COBA models operate in different parameter regimens.

With sufficiently strengthened pathway synapses, a rate-coded signal can travel through at least six synaptic stages with a transmission delay of ~20 ms. A 13-fold increase in synapse strength was needed to optimally transmit signals through a network. This corresponds to evoked EPSPs, within the active network, of ~8 mV, which is relatively large but not unheard of (Song et al., 2005). In the 10,000 neuron, randomly connected networks we studied, propagation involved only three pathway synapses. One way to get more synapses involved in the signal propagation and, thus, to lessen their strength, is to use a structured architecture, more like the real cortex, with columns and targeted branching axons. This is a topic of ongoing research.

The optimal synapse enhancement depends on the nature of the transmitted signal, especially in the CUBA model. The critical factors seem to be the ratio of silent to active periods and the distribution of firing rates within the active signal. A signal with long silences or low firing rates requires a smaller optimal synapse enhancement than one with large amounts of high-frequency

E53 XOR 门的定量行为
作者报告｜四种计算单元中唯一给出定量判据的结果：单输入 > 60 Hz、双输入同时激活 < 40 Hz。作者说明两个输入层分别驱动各开关的兴奋与抑制中间细胞，因此同时输入会互相压制。

E56 触发器失败的具体机制
作者报告｜把 E54 的失败归因到具体环节：一个环路的抑制输出有时不足以静默另一个环路。这与作者在 Discussion 中的总结一致——噪声是逻辑门正常工作的必要条件，也是它们偶尔失败的原因。

E54 触发器不是完美的记忆
作者报告｜记忆单元的失败模式。作者把它归因于背景活动涨落导致抑制不足，并指出模拟约 800 ms 时即出现。这一条同时说明本文的高发放率维持依赖背景活动，而背景活动也会破坏它。

E64 噪声的非可调性作为贡献
作者报告｜作者强调相对早期工作的改进在于噪声来自网络内部且不可自由调整，从而必须让网络参数自行适配噪声。这一条界定了本文贡献的真正内容：不是把噪声调好，而是让系统适应固定噪声。

E65 COBA 定量优于 CUBA
作者报告｜作者对两个模型的比较结论，注意其限定是「无需重新调参即可传输更宽的速率范围」，而不是 COBA 在任何条件下都更优。

E66 对 COBA 优势的归因及其张力
作者报告｜作者把 COBA 的优势归因于电导机制本身而非参数区域差异。但该主张与 p. 5「只有 COBA 能在高突触强度区域给出合理结果」以及 p. 9 承认在低电导区域测试结论相似之间存在张力，因此不宜把这条读作已排除参数差异。

Ｅ５５　　四种电路的共同缺陷：响应迟缓
作者报告｜作者自己给出的横向限制：四种计算单元响应都偏慢，且被更长的抑制性突触时间常数进一步放大。这与步骤　３　中频率响应窄的结论同源，说明开关式电路不适合快速时序信息处理。

Ｅ５８　　讨论中给出的传播能力总述
作者报告｜作者在　Ｄｉｓｃｕｓｓｉｏｎ　汇总的传播能力：至少六个突触级、约　２０　ｍｓ　延迟。注意作者用「ａｔ　ｌｅａｓｔ」而非给出上限，六层的限制来自架构与搜索规则（见　Ｅ０９、Ｅ１０）。

Ｅ５９　　最优强度的汇总数字
作者报告｜１３　倍这一数字在摘要之外被再次确认，可作为引用时的主数值。应注意它对应的是　ＣＯＢＡ　模型中平台区的最优因子　１２（因子定义为相对非通路突强度的增量）。

Ｅ６０　ＥＰＳＰ　幅度的合理性声明
作者报告｜作者为　８　ｍＶ　这个偏大的幅度给出经验参照（引　Ｓｏｎｇ　等　２００５）。这是把模型参数与实验数据挂钩的唯一一处，但属定性类比，本文未做定量比较。

Ｅ６１　结构性弱点：每层只用到　３　条突触
作者报告｜作者自己指出结论的结构性弱点：传播只依赖极少的突触，因此对单条突触强度的改变格外敏感。这也是必须做突触因子扫描、且扫描窗口较窄的根本原因。

Ｅ６２　　改进方向：结构化架构
作者报告｜作者提出的未来方向：用皮层柱与靶向分支轴突让更多突触参与传播，从而降低单条强度。这明确表明本文的稀疏随机架构被作者自己视为一个需要改进的简化。

Ｅ５７　　模拟调制：只有断言没有数据
作者报告｜作者紧接着注明「ｄａｔａ　ｎｏｔ　ｓｈｏｗｎ」。这是本文唯一一处无证据支持的功能主张，引用时必须标明该结果在原文中未展示。

firing. The poorer performance of the CUBA model may be attributable to the fact that the membrane potential for CUBA neurons hovers close to the threshold, which increases the propagation of ghost signals, as well as the decreased membrane time constant in the COBA model.

Signal propagation in the networks studied falls off as a function of frequency much as it does in the real cortex (Movshon et al., 1978; Orban et al., 1985; Hawken et al., 1996). The failure of the network to transmit high-frequency signals is the root of its problems in supporting synfire waves. Although the probability to evoke a single postsynaptic response to an input rises linearly with the number of presynaptic spikes, the probability of evoking a second spike increases as well. Thus, synapses sufficiently strong to propagate the leading edge of a synfire wave create a wake of secondary spikes. Achieving synfire propagation requires a fine balance between noise and synaptic strength. For synfire chain propagation to succeed in the network we studied, gain modulation had to be used to decrease the effect of noise along the pathway. The parameter regimen over which synfire propagation is possible is rather small.

A useful way to think about propagation is as an avalanche in which each presynaptic neuron triggers activity in a postsynaptic neuron with a certain probability (Harris, 1963; Zapperi et al., 1995; de Carvalho and Prado, 2000; Beggs and Plenz, 2003). A critical cascade occurs when, on average, each presynaptic neuron fires one of its postsynaptic targets. In the pathways we consider, the average number of synaptic targets of pathway neurons is three, so we would expect the probability that a presynaptic spike generates a postsynaptic spike in any one of these targets to be 0.33 at criticality. Figure 5*d* shows that the spiking probability is ~0.25 at the optimal synaptic strength. This is somewhat lower than the critical value; even synchronous stimulation with three spikes evokes a postsynaptic spike only 80% of the times. The reason for this discrepancy is the probability of evoking a secondary spike (Fig. 6*f*,*g*). The added probabilities for primary and secondary spikes are ~0.3, close to the critical value. Although this does not take into consideration the temporal slurring that must occur as a result of delivering 20% of the stimulus 5 ms late, it is reassuring that these numbers are fairly close to criticality, guaranteeing spike number conservation.

The existence of multiple pathways in a network introduces the possibility of gating interactions between them. Finding candidate pathways makes it possible to create logic gates, switches, and memory units by strengthening selective synapses. Noise was required for the logic gates to function properly, but it also caused them to fail sometimes. Remarkably, all of these circuits could be found within the random network, so only synaptic enhancement was required to make them functional. We did this strengthening by hand, but it would be interesting to investigate whether various synaptic plasticity mechanisms could do this autonomously.

The problem we address, how to get signals over a wide dynamic range to propagate reliably across neural circuits, is an important element for understanding any sort of brain function. We showed that networks based on our understanding of background activity provide a sufficient source of noise to support signal propagation and that useful computations can be performed by interacting pathways found within such networks.

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E39 synfire 在原始网络中失败
作者报告｜synfire 的基线与 rate 相同：不过第 1 层。作者指出这与早期文献「远多于三个 spike 单位才可能成功」的结论一致，说明失败是稀疏连接的必然。

E70 synfire 必须靠增益调制压制噪声
作者报告｜作者对必要条件的总结。值得注意的是增益调制在此处的作用是压制噪声而非放大信号，与步骤 3 中它对 rate 传播无效的结论并不矛盾——两者说的是同一操作在不同目的下的效果。

E71 synfire 的参数区域相当小
作者报告｜作者对 synfire 脆弱性的最终判断。引用时注意原文用词是 regimen 而非 regime（这是本次校对发现的一处常见转写错误），且「相当小」是定性判断，本文未给出参数范围的数值边界。

E67 雪崩类比给出的临界预期
作者报告｜作者用临界级联理论给出独立预期：3 个突触靶点在临界时应对应 0.33 的单突触后发放概率。这是本文唯一的理论性自洽性检查，其价值在于可证伪而非已被独立验证。

E68 实测值低于临界预期
作者报告｜实测 0.25 低于理论预期 0.33。作者的解释是二者之差来自二级发放概率（作者称一级加二级约 0.3），但也承认未考虑 20% 刺激延迟 5 ms 带来的时间涂抹，因此该吻合是近似的。

E73 电路存在于随机网络中
作者报告｜作者对可行性的最强表述。注意这里的「found」指按预写连接条件搜到，因此支持的是「无需新建连接」，不支持「电路会自行涌现」。

E74 作者点名的未解决问题：可塑性
作者报告｜作者唯一明确写出的未来工作：由某种可塑性机制自主完成手工所做的 13 倍选择性增强。这也是本文最主要的未实现环节——所有突触调整都是外部给定的，因此学习部分完全缺失。